MH EMCR AI Workshop: Embracing not avoiding AI to enhance science teaching and writing

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The Challenge – A changing landscape

Modernizing assessment in Cell and Molecular biology

A real-world test of artificial intelligence infiltration of a university examinations system: A "Turing Test" case study

Peter Scarfe, Kelly Watcham, Alasdair Clarke, Etienne Roesch

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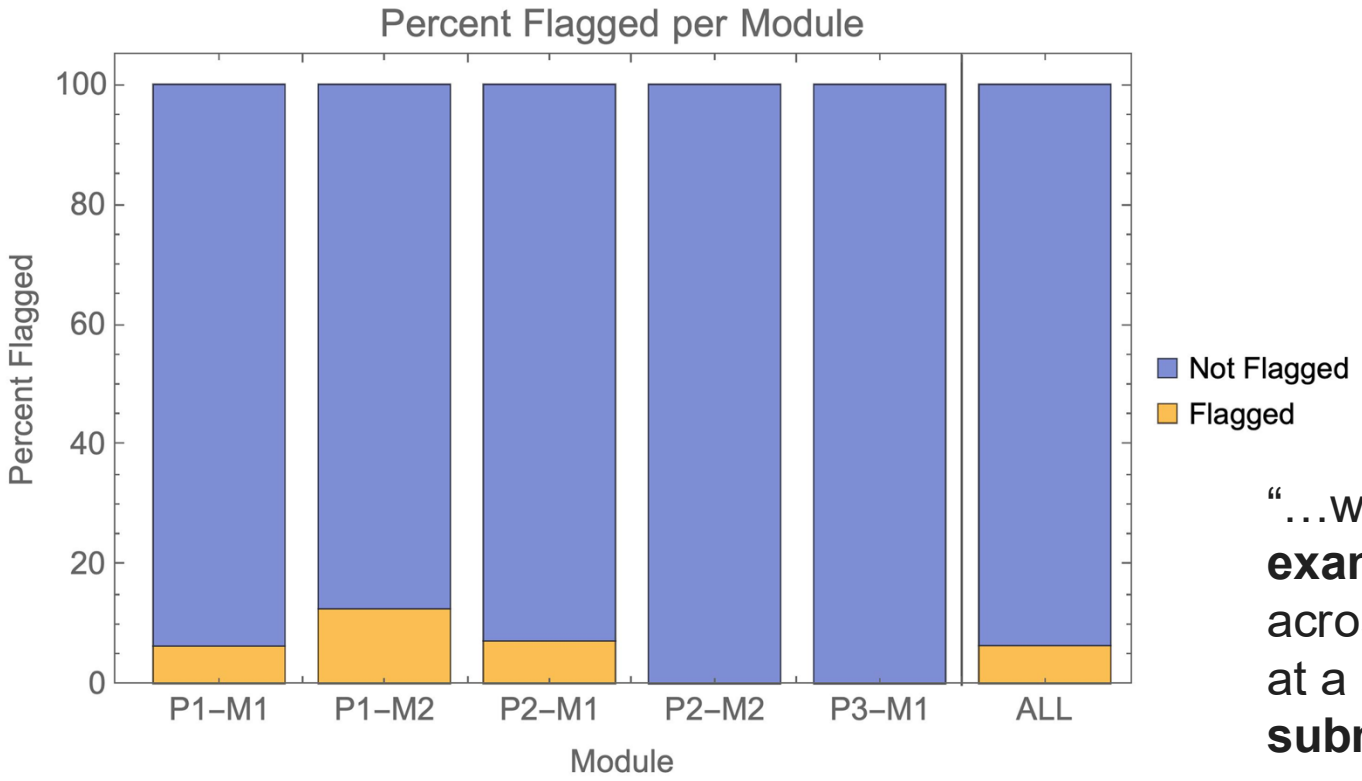


- **Generative AI is here to stay**
- **Students are using it**
- **Studies show we are unable to detect its use!**

"…we injected **100% AI written submissions into the examinations system** in five undergraduate modules, across all years of study, for a BSc degree in Psychology at a reputable UK university. We found that **94% of our AI submissions were undetected."**

The Challenge – A changing landscape

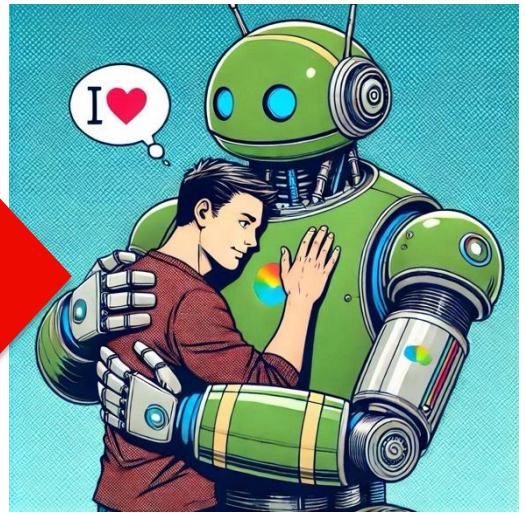
Modernizing assessment in Cell and Molecular biology

- Computational skills including using AI are now an essential graduate skill
- We still must provide our core education in biology, practical skills and communication
- **Large cohorts constrain writing assessments** – particularly in marking
- **Biology Subjects requires redesign** to provide career relevant graduate skills

Our Aim

Redesigning Biol340

Adopt an ***embrace not avoid*** mindset for GenAI in higher education



- Update the content and assessments in Biol340 to include skills in data analysis, communication and GenAI use
- Reduce the number and length of wet labs to enable new tutorial style dry labs on data analysis and Gen AI use
- Adapt our assessments to include generative AI as a core component
- Re-introduce scientific writing assessment

The Challenge – Past Biol340 Structure

Modernizing assessment in Cell and Molecular biology

Assessment

- Weekly Lectures (2hr/block)
  - Three theory quizzes (65% of total)
- Five practical wet labs (4hr/lab)
  - Five pre-lab quizzes (10% of total)
  - Five post-lab quizzes (25% of total)

- Class size of ~200 students most with a molecular biology background
- Student feedback that labs are too long with long period of "downtime"
- Data analysis takes place during labs but is often at the end of the classes without guided tutorial
- Current assessment model relies entirely on quizzes
- No education or assessment on scientific writing or communication
- No GenAI education

Redesigning Biol340 – New structure

Incorporating GenAI & computer literacy into teaching scientific writing

- Weekly Lectures (2hr/block)
  - Mid-session quiz (20% of total)
  - Final Exam (30% of total)
- Four practical wet labs (3hr/lab)
  - Four post-lab quizzes (20% of total)
  - Peer assessed scientific report (15% of total)
- Four dry labs (2hr/lab)
  - Generative AI & peer-review reflection task (15% of total)

- One wet lab removed and all shortened by 1 hr
- Four new dry labs introduced on topics:
  - Data analysis in R
  - Scientific writing
  - Peer review and feedback
  - GenAI literacy
- Two new assessments framed around a scientific report and review introduced

Redesigning Biol340 – Peer-assessed scientific writing

New Assessments

- Working in pairs to develop a piece of formal scientific writing
- Utilise data analysis skills & coding from dry pracs
- Reports are peer marked following established rubric and written feedback is given
- Peer-marking is individual
- Student sees a variety of report and can reflect on writing/communication
- Report then serves as draft for reflection assignment

Using **peer-assessment** to help inform future writing

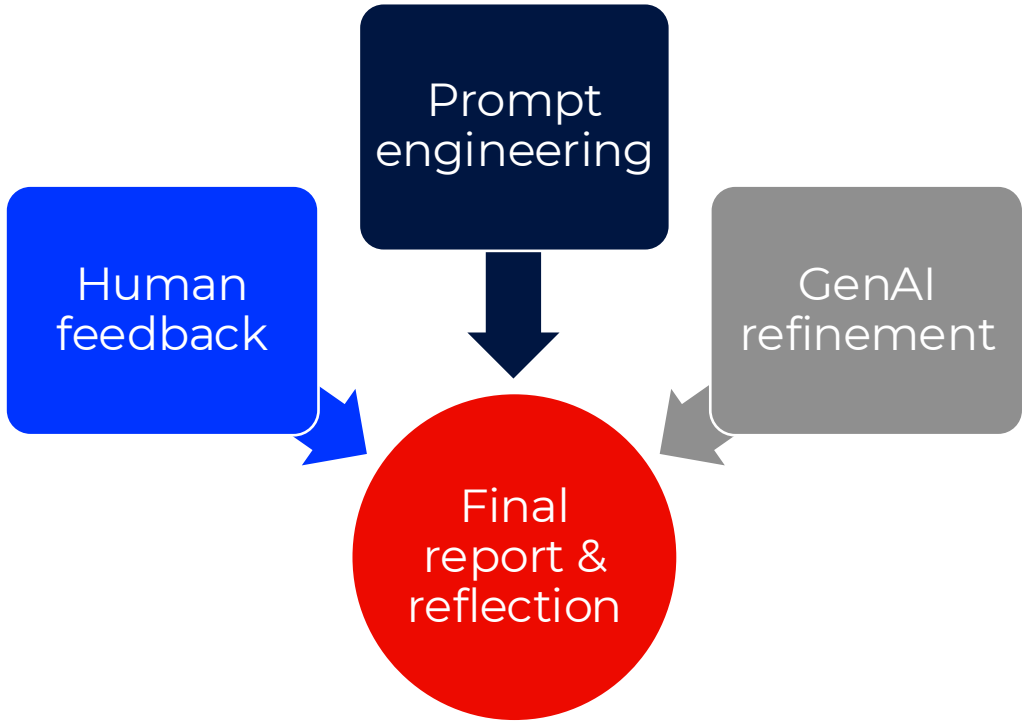
Redesigning Biol340 – Generative AI reflection piece

New Assessments

Builds of human feedback provided during peer-assessment

Using a template – students individually:

- Summarise the human feedback on their writing
- Completes a revision guided through GenAI refinement of writing using increasing levels of 'prompt complexity'
- compare their drafts pre- and post-refinement and discuss how AI changed their work
- Reflect on human vs. AI feedback and the use of AI tools for scientific writing



GenAI assessment example

Part 2: Revising Your Report Using AI
Instructions: Experiment with the following three levels of prompts while revising your report. Document your prompts, AI outputs, and how you applied the feedback.

Prompting Log
2.0. Contextualise the model

- **Prompt used:**
  (e.g., "You are an expert in scientific writing and an editor of a top ranked journal")

2.1. "Do for me" Prompt

- **Prompt used:**
  (e.g., "Rewrite my discussion section to make it clearer.")

- **AI Output:**
  (Paste the relevant output here)

By guiding through GenAI use we hope to:

- Inform better GenAI use
- Teach prompt engineering fundamentals
- Minimise unethical AI use (i.e. direct copy-pasting without attribution)
- Improve confidence in scientific writing & editing

GenAI Reflection – AI vs Human Feedback

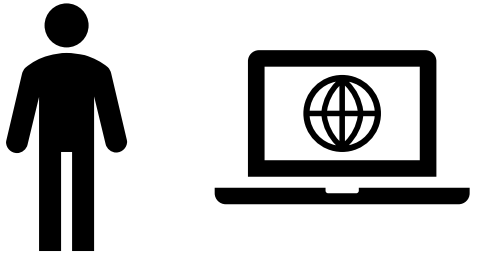
AI feedback was clearer, more actionable for writing structure and tone

Human feedback valued for context and empathy

“AI gave me feedback on how to improve flow and grammar — it was more helpful than the vague ‘good job’ from my peer.”

“I liked that the AI could pinpoint sentence structure issues and suggest alternatives on the spot.”

“The peer knew what we were meant to do — AI sometimes gave feedback that didn’t match the report format.”



GenAI Reflection – Writing Improvement

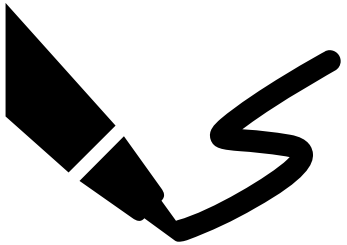
AI helped polish writing and reduce redundancy

Students reported better structure, clarity, tone after AI refinement

“AI helped me realise I was being repetitive — I rewrote the discussion to focus more on results interpretation.”

“I’m more confident in writing scientific paragraphs after seeing how to link ideas better.”

“Now I notice when my paragraphs lack direction. I’ll ask the AI to help make them flow logically.”



GenAI Reflection – Prompting Strategies

Poor prompts gave generic answers

Students found higher complexity prompts gave better responses

“When I just said ‘fix this paragraph’, the response was bland. But when I asked it to ‘critique like a biology tutor’, it worked better.”

“The better I got at prompting, the better the AI became at helping me improve.”

“Prompting was trial and error — but once I figured it out, the feedback got way better.”

A short primer on Prompt Engineering

How we equipped our students to be AI "super users"

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Prompt engineering

Levels of complexity – getting the most from your prompts

1) “Do”

“Make me a scatter plot from this data."

2) “Evaluate”

"Evaluate the code I have written."

3) “Teach”

"Teach me how to make visually appealing graphs in ggplot2."

4) Model contextualisation

“You are an expert in data visualisation and an programmer in R language, give me feedback ….”

Prompt engineering

Levels of complexity – getting the most from your prompts

1) “Do”

Today

Write me code in R that will create a scatter plot for data on cell viability with these columns: (Absorbance, Dillution, Treatment type)

Copilot

Sure! Here's an example of how you can create a scatter plot in R using the `ggplot2` package. This code assumes you have a data frame named `cell_data` with columns `Absorbance`, `Dillution`, and `Treatment_type`.

```
# Load the ggplot2 package
library(ggplot2)

# Sample data frame
cell_data <- data.frame(
  Absorbance = c(0.1, 0.2, 0.3, 0.4, 0.5),
  Dillution = c(1, 2, 3, 4, 5),
  Treatment_type = c("A", "B", "A", "B", "A")
)

# Create the scatter plot
ggplot(cell_data, aes(x = Dillution, y = Absorbance, color = Treatment_type)) +
  geom_point(size = 3) +
  labs(title = "Scatter Plot of Cell Viability",
       x = "Dillution",
       y = "Absorbance") +
  theme_minimal()
```

This code will create a scatter plot with `Dillution` on the x-axis, `Absorbance` on the y-axis, and different colors for each `Treatment_type`. You can customize the plot further by adjusting the parameters in the `ggplot` function.

Prompt engineering

Levels of complexity – getting the most from your prompts

4) “teach”

Today

Teach me how to write an effective R code to create a scatter plot for data on cell viability with these columns: (Absorbance, Dillution, Treatment type)

Copilot

4. **Customize your plot** (optional): You can further customize your plot by adding more elements or adjusting the theme. For example:

```
ggplot(data, aes(x = Dillution, y = Absorbance, color = Treatment.type)) +
  geom_point(size = 3, alpha = 0.7) +
  labs(title = "Scatter Plot of Cell Viability",
       x = "Dillution",
       y = "Absorbance",
       color = "Treatment Type") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```

This code will create a scatter plot with `Dillution` on the x-axis, `Absorbance` on the y-axis, and different colors for each `Treatment.type`. You can adjust the size and transparency of the points with `size` and `alpha` parameters in `geom_point()`.

Feel free to ask if you have any questions or need further customization!

The importance of contextualisation

You are a primary school student, write me an abstract for a project on mitochondrial diseases

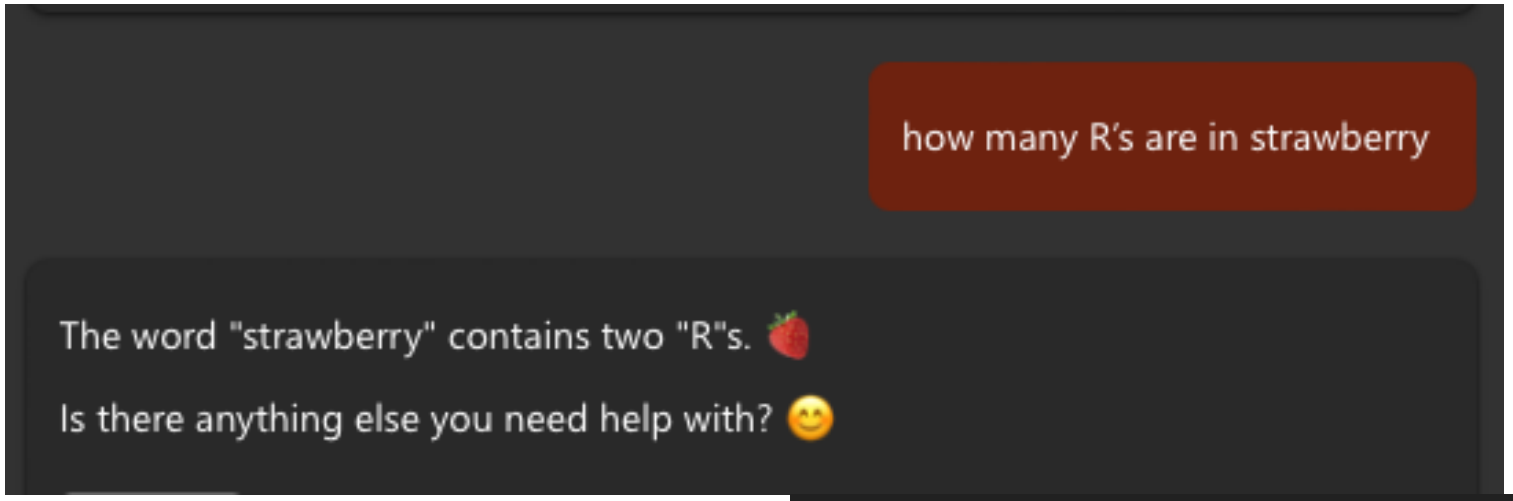
Mitochondrial diseases are when parts of our body don't work right because the **little "batteries"** inside our cells, called mitochondria, **get sick**

You are a senior lecturer and expert in biochemistry and cell and molecular biology. Write me an abstract about mitochondrial diseases

Mitochondrial diseases are a diverse group of disorders resulting from **dysfunction in** the mitochondria, **the energy-producing organelles in eukaryotic cells**. These diseases often present with multisystemic manifestations, particularly affecting high-energy-demand tissues such as the brain, muscles, and heart.

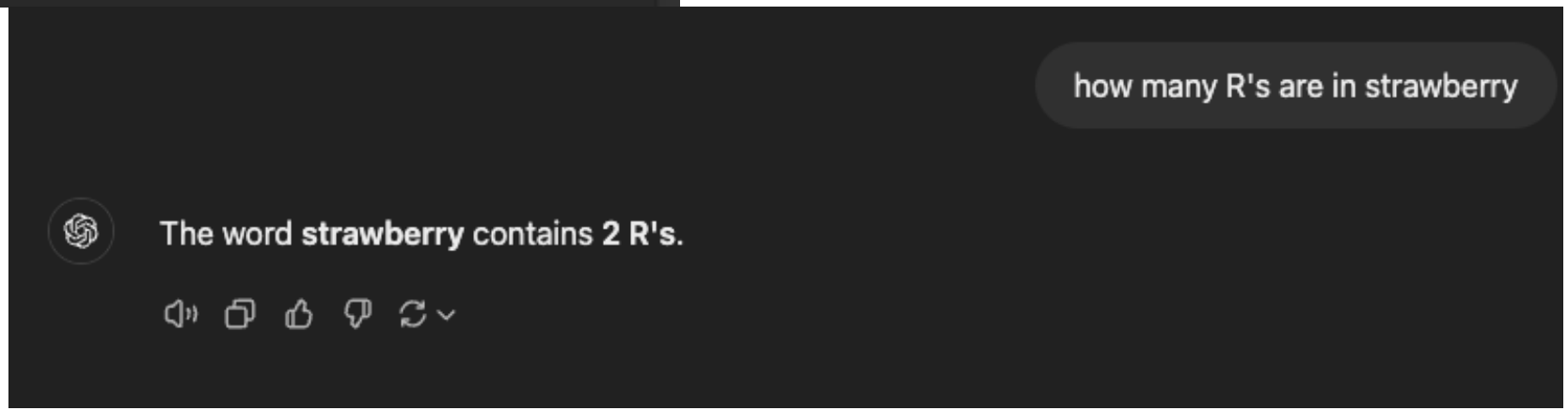
AI is a powerful tool – if contextualized correctly

Prompt engineering and training essential!



Co-pilot

chatGPT-4



AI is a powerful tool – if contextualized correctly

Prompt engineering and training essential!

chatGPT-4

You are an educator in linguistics with expertise in writing. Evaluate how many R's there are in strawberry

As a linguist and writer, let's break this down systematically:

The word **strawberry** contains the following sequence of letters: **s-t-r-a-w-b-e-r-r-y**. Now, let's count the **R's**:

1. The first **R** appears in the third position (**st→r**).
2. The second **R** appears in the eighth position (**be→r**).
3. The third **R** appears in the ninth position (**r→y**).

Thus, **strawberry** contains **3 R's**.

This method ensures accuracy by verifying each occurrence systematically.

A quick guide to a practical use of AI in teaching

or how to save hours of time with a little bit of problem solving

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Some practical uses of AI in teaching

Problems can be easily solved with AI if you reframe as a coding/computation program

Examples:

- Making an online moodle quiz from scratch, or converting a paper based quiz to moodle.
- Developing a website to teach data analysis
- Preparing a powerpoint scaffold for a new tutorial.

Some practical uses of AI in teaching

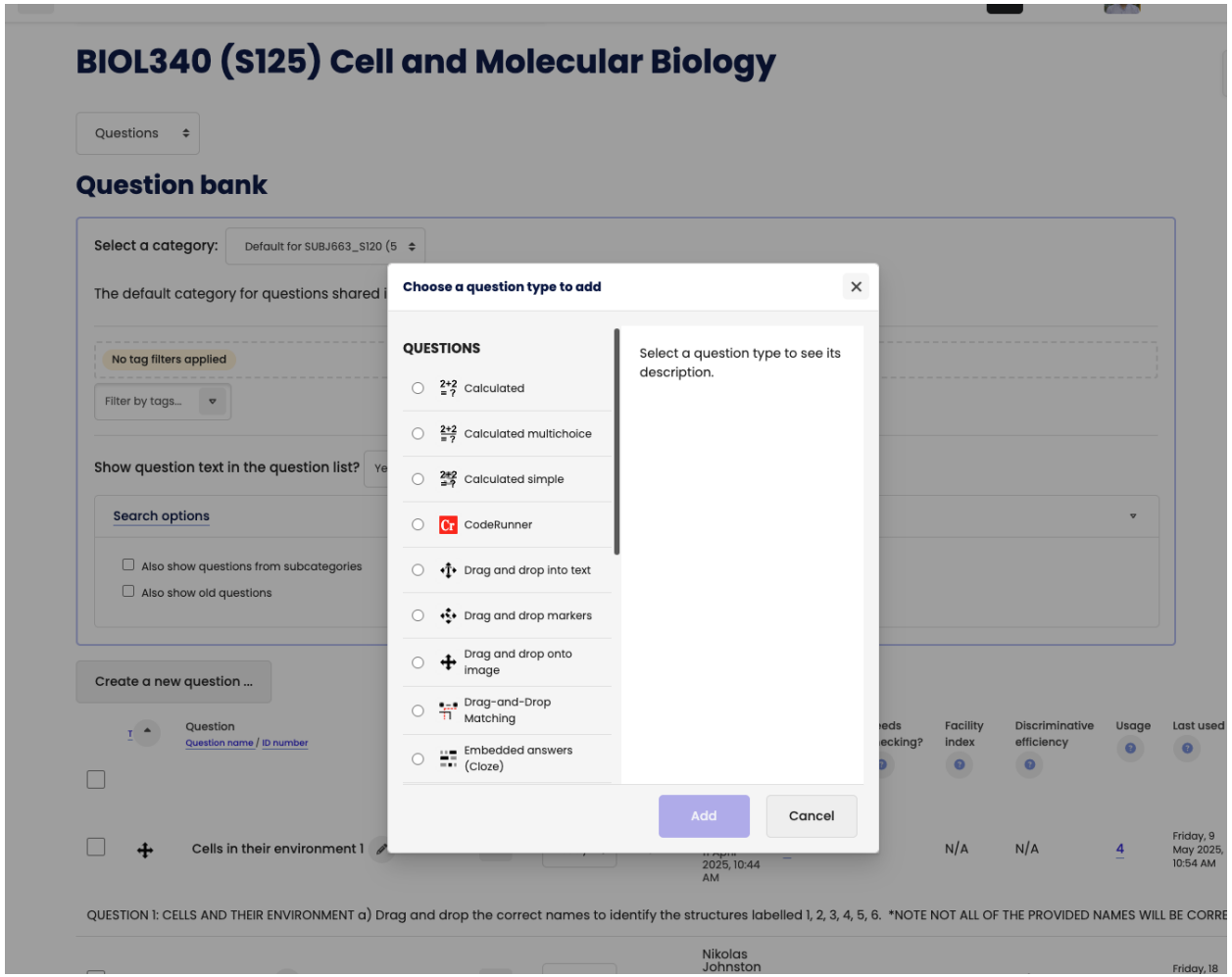
Problems can be easily solved with AI if you reframe as a coding/computation program

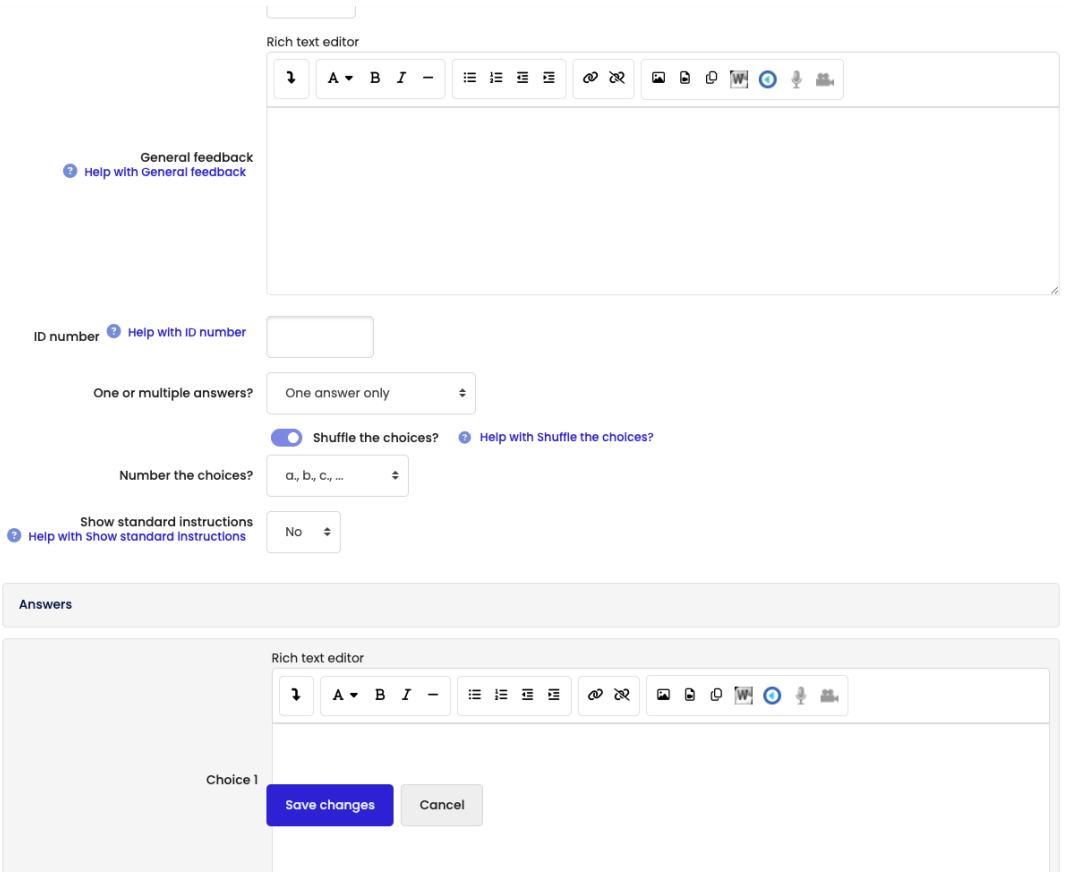
Examples:

- **Making an online moodle quiz from scratch, or converting a paper based quiz to moodle.**
- Developing a website to teach data analysis
- Preparing a powerpoint scaffold for a new tutorial.

The hard way ☹ – Manually enter each question

Graphical user interfaces (GUI) can be the enemy. Here we need to select question type, manually enter the text and put in all the correct answers





The less hard way 😐 –Use cloze style

We can do better by choosing cloze type which at least allows us to have inline questions and answers– but you have to know the cloze language which is a barrier to use.

General

Current category **Current category** Default for SUBJ663_S120 (57)

Version Version 1
Created by *Nikolas Johnston* on *Friday, 4 April 2025, 11:28 AM*

Question name Required

functions 1

Rich text editor

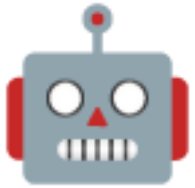
Question text Required

If I want to add a new column to a dataframe called "bugs" that was called "flies" in R using the methods you have learned in the pre-lab, how could I arrange the code?

{1:MULTICHOICE_S:~%100%bugs~%0%flies~%0%mutate~%0%filter~%0%select~%0%rename} <-
{1:MULTICHOICE_S:~%0%bugs~%0%flies~%100%mutate~%0%filter~%0%select~%0%rename}
({1:MULTICHOICE_S:~%100%bugs~%0%flies~%0%mutate~%0%filter~%0%select~%0%rename},
{1:MULTICHOICE_S:~%0%bugs~%100%flies~%0%mutate~%0%filter~%0%select~%0%rename})

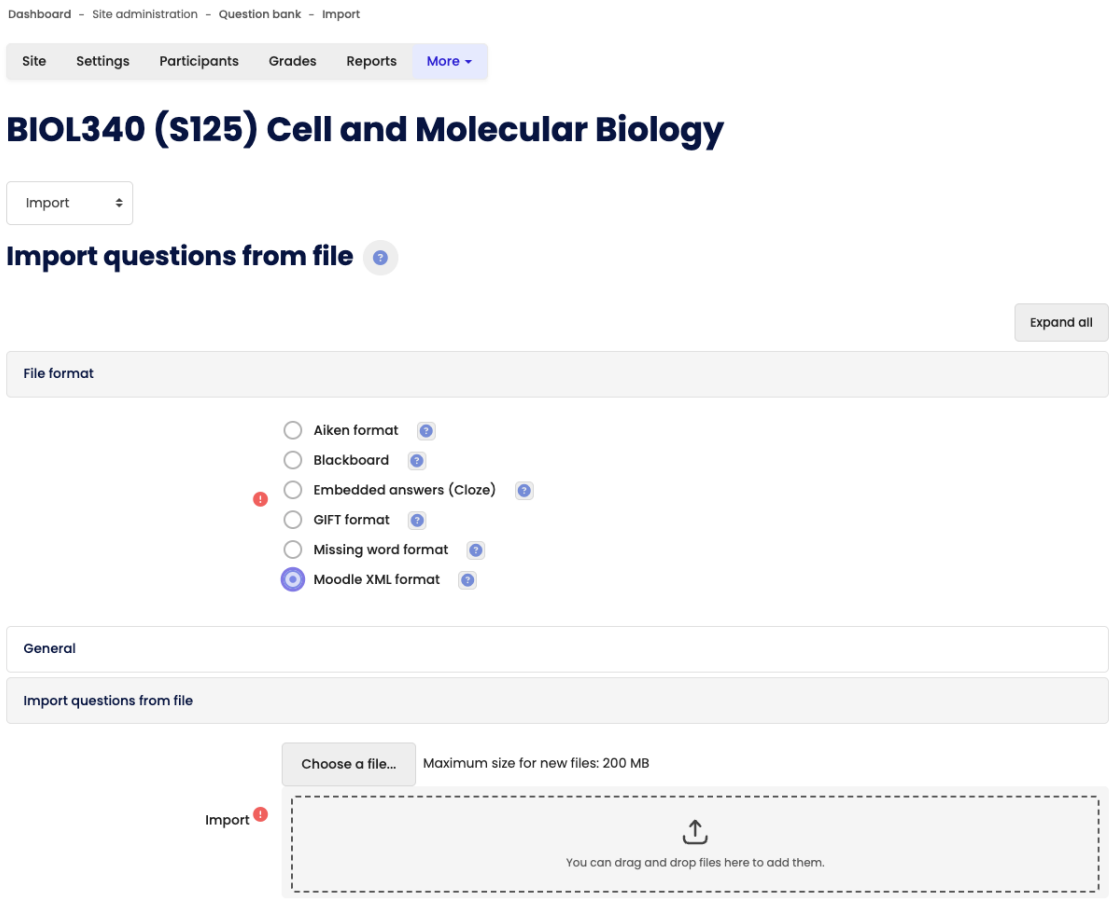
Question status Ready

The AI enhanced way

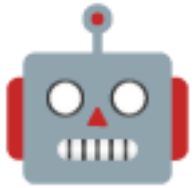


Find out how Moodle stores questions —— in this case as an XML file

See if there is a way to import —— Yes there is!



The AI enhanced way



Now let's write an exam! (keep in mind you need to proof everything)

Option 1: feed in a list of pre existing questions and convert to moodle cloze style as an XML.

Option 2: provide a PDF of a paper based quiz and ask moodle to convert into cloze style and provide the whole quiz as an XML

You are a university curriculum developer working with a biology course and making moodle quizzes.

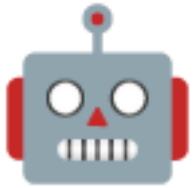
take the following true false question text and covert it to a Moodle XML file for upload:

True or False: Motor proteins use ATP hydrolysis to drive conformational changes that enable movement along cytoskeletal filaments.
→ True
True or False: All motor proteins can move bidirectionally along their cytoskeletal tracks depending on the ATP concentration.
→ False

True or False: The movement of motor proteins is powered by a repeated cycle involving ATP hydrolysis, ADP release, and ATP rebinding.
→ True
True or False: Kinesin, dynein, and myosin share common structural features despite interacting with different cytoskeletal elements.
→ True
True or False: Motor proteins are capable of moving along a cytoskeletal filament for only a single step before detaching.
→ False

True or False: The motility systems in cells function by converting chemical energy from ATP into mechanical work through motor proteins.
→ True

The AI enhanced way



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you are a higher education curriculum development specialist advising me on a cell biology subject for thirsd year students. I have this question for a final exam., I want to convert it to a moodle cloze style can you do that for me

Absolutely — please share the exam question you'd like to convert into a Moodle **Cloze** question format, and I'll turn it into the appropriate Moodle XML syntax for upload.

Let me know if you'd prefer:

- **Dropdown (multiple-choice)**
- **Short answer (typed response)**
- **Numerical (with tolerance)**

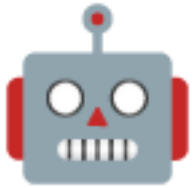
Also, let me know if you'd like **feedback** or **hints** included for students.

Question 1 - Cell Cycle

a) Name the four phases of the cell cycle and briefly describe one event that occurs in each of these phases (2 marks)

Thanks! Here's a **Moodle Cloze question** version of your **Question 1a** in **dropdown style**, suitable for third-year cell biology students. This format uses embedded answers with short feedback (optional and editable):

The AI enhanced way



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The AI enhanced way

```
::Q1a - Cell Cycle Phases and Events::

Match each phase of the cell cycle with a key event that occurs during it. Each
dropdown contains one correct option.


**G1 phase**: {1:MC:=Cell grows and performs normal functions ~DNA is replicated
~Chromosomes align at the metaphase plate ~Cell divides into two daughter cells}


**S phase**: {1:MC:=DNA is replicated ~Cell grows and performs normal functions
~Chromosomes condense ~Spindle fibres form}


**G2 phase**: {1:MC:=Cell prepares for mitosis ~DNA is replicated ~Chromosomes separate
~Cytokinesis begins}


**M phase**: {1:MC:=Cell divides into two daughter cells ~DNA is replicated
~Centrosomes duplicate ~Cell grows and performs normal functions}
```

The AI enhanced way

Can you give me this as a moodle xml file

Analyzed

python

Always show details

Copy

```python
from xml.etree.ElementTree import Element, SubElement, tostring, ElementTree
import xml.dom.minidom as minidom
from pathlib import Path

# Create the root <quiz> element
quiz = Element('quiz')

# Create question element
question = SubElement(quiz, 'question', type="cloze")

# Question name
name = SubElement(question, 'name')
text = SubElement(name, 'text')
text.text = "Q1a – Cell Cycle Phases and Events"

# Question text
questiontext = SubElement(question, 'questiontext', format="html")
qt_text = SubElement(questiontext, 'text')
qt_text.text = (
    "<![CDATA["
    "<p>Match each phase of the cell cycle with a key event that occurs during it. Each d
    "<p><strong>G1 phase</strong>: {1:MC:=Cell grows and performs normal functions ~DNA i
    "<p><strong>S phase</strong>: {1:MC:=DNA is replicated ~Cell grows and performs norma
    "<p><strong>G2 phase</strong>: {1:MC:=Cell prepares for mitosis ~DNA is replicated ~C
    "<p><strong>M phase</strong>: {1:MC:=Cell divides into two daughter cells ~DNA is rep
```

Here is your Moodle XML file for **Question 1a – Cell Cycle Phases and Events:**

Download cell_cycle_cloze.xml

You can upload this file directly to your Moodle question bank or quiz using the **Moodle XML format** option. Let me know if you'd like additional questions or a version using typed short answers instead of dropdowns.

Import

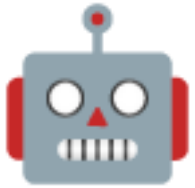
Parsing questions from import file.

Importing 20 questions from file

1.

2.

The AI enhanced way



Question name / ID number				by First name / Last name / Date	
MC-20. Shared DNA	Edit	Ready	v1	Nikolas Johnston 1 August 2025, 12:05 PM	0
MC-01. Largest Land Mammal	Edit	Ready	v1	Nikolas Johnston 1 August 2025, 12:05 PM	0
MC-02. City in Brazil	Edit	Ready	v1	Nikolas Johnston 1 August 2025, 12:05 PM	0
MC-03. Jerusalem Artichoke	Edit	Ready	v1	Nikolas Johnston 1 August 2025, 12:05 PM	0

The AI enhanced way – Takeaway

It very valuable to know the file types and computer language of the programs and websites you are interfacing with

Try and reframe your problem in a way AI is good at answering!

Questions?

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